

Вед №13 Роряция Тейлора.

при $x \rightarrow 0$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\operatorname{tg} x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

$$\operatorname{arcsin} x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$$

$$\operatorname{arctg} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$(1+x)^a = 1 + ax + \frac{a(a-1)}{2!}x^2 + \frac{a(a-1)(a-2)}{3!}x^3 + \dots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots$$

$$f(x) = (1+x)^a$$

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k + o(x^n)$$

$$(1+x)^a = \sum_{k=0}^n \frac{a(a-1)(a-2)\dots(a-k+1)}{k!} x^k + o(x^n)$$

$$(1+x)^a = 1 + ax + \frac{a(a-1)}{2!}x^2 + \frac{a(a-1)(a-2)}{3!}x^3 + o(x^3)$$

k	$f^{(k)}(x)$	$f^{(k)}(0)$
0	$(1+x)^a$	1
1	$a(1+x)^{a-1}$	a
2	$a(a-1)(1+x)^{a-2}$	$a(a-1)$
3	$a(a-1)(a-2)(1+x)^{a-3}$	$a(a-1)(a-2)$
...	...	...
k	$a(a-1)(a-2)\dots(a-k+1)(1+x)^{a-k}$	$a(a-1)(a-2)\dots(a-k+1)$

$$\sqrt{1+x} = (1+x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + o(x^3)$$

Пример 1

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} = \lim_{x \rightarrow 0} \frac{x - \frac{x^3}{6} + o(x^3) - x}{x^3} = \lim_{x \rightarrow 0} \underbrace{-\frac{1}{6}}_{-\frac{1}{6}} + \lim_{x \rightarrow 0} \underbrace{\frac{o(x^3)}{x^3}}_{0 \text{ по опр } o\text{-малости}} = -\frac{1}{6} + 0 = \boxed{-\frac{1}{6}}$$

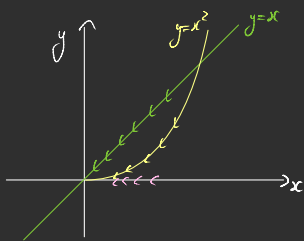
$$\sin x = x - \frac{x^3}{6} + o(x^3)$$

$$f(x) = o(g(x)) \text{ при } x \rightarrow x_0,$$

$$\text{если } \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0 \quad // f(x) - \text{д.м. по сравнению с } g(x) //$$

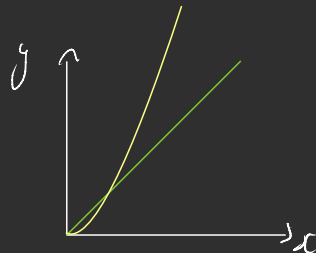
$$x^2 = o(x) \text{ при } x \rightarrow 0 \quad (x^2 \text{ д.м. по ср. с } x \text{ при } x \rightarrow 0)$$

$$\lim_{x \rightarrow 0} \frac{x^2}{x} = \lim_{x \rightarrow 0} x = 0 \quad \checkmark$$



$$x = o(x^2) \text{ при } x \rightarrow \infty \quad (x \text{ д.м. по ср. с } x^2 \text{ при } x \rightarrow \infty)$$

$$\lim_{x \rightarrow \infty} \frac{x}{x^2} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \quad \checkmark$$



$$\text{При } x \rightarrow 0 \quad o(x^n) + o(x^m) = o(x^{\min(n,m)})$$

$$\text{При } x \rightarrow \infty \quad o(x^n) + o(x^m) = o(x^{\max(n,m)})$$

$$x \rightarrow 0 \quad o(x^2) + o(x^3) = o(x^{\min(2,3)}) = o(x^2)$$

$$x \rightarrow \infty \quad o(x^2) + o(x^3) = o(x^{\max(2,3)}) = o(x^3)$$

$$x^{2.5} = o(x^2) \text{ при } x \rightarrow 0$$

$$\lim_{x \rightarrow 0} \frac{x^{2.5}}{x^2} = \lim_{x \rightarrow 0} x^{0.5} = 0$$

$$x^{2.5} = o(x^3) \text{ при } x \rightarrow 0$$

$$\lim_{x \rightarrow 0} \frac{x^{2.5}}{x^3} = \lim_{x \rightarrow 0} \frac{1}{x^{0.5}} = \frac{1}{0} \neq 0$$

$$o(x) + o(x) = o(x)$$

$$o(x) - o(x) = o(x)$$

$$x^n \cdot o(x^m) = o(x^{n+m})$$

$$C \cdot o(x^n) = o(x^n)$$

$$o(x^n) \cdot o(x^m) = o(x^{n+m})$$

Пример 2

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{1 + x + \frac{x^2}{2} + o(x^2) - 1 - x}{x^2} = \lim_{x \rightarrow 0} \left(\frac{1}{2} + \frac{o(x^2)}{x^2} \right) = \left(\frac{1}{2} \right)$$

$\downarrow$ $\downarrow$
 $\frac{1}{2}$ 0

$$e^x = 1 + x + \frac{x^2}{2} + o(x^2)$$

Пример 3

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3} = \lim_{x \rightarrow 0} \frac{\frac{x^3}{6} + o(x^3)}{x^3} = \left(\frac{1}{6} \right)$$

$\frac{x^3}{6} + o(x^3)$
 $\uparrow$ $\uparrow$
 $\frac{1}{6}$ 0

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

Пример 4

$$\lim_{x \rightarrow 0} \frac{\cos x - e^{-x^2/2}}{x^4} = \lim_{x \rightarrow 0} \frac{1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4) - 1 + \frac{x^2}{2} - \frac{x^4}{8} + o(x^4)}{x^4} = \lim_{x \rightarrow 0} \frac{-\frac{1}{12}x^4 + o(x^4)}{x^4} \xrightarrow{0} -\frac{1}{12}$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$

$$e^x = 1 + x + \frac{x^2}{2} + o(x^2)$$

$$e^{-\frac{x^2}{2}} = 1 - \frac{x^2}{2} + \frac{x^4}{8} + o(x^4)$$

Пример 5

$$\lim_{x \rightarrow 0} \frac{e^x \sin x - x(1+x)}{x^3} = \lim_{x \rightarrow 0} \frac{x + x^2 + \frac{x^3}{3} + o(x^3) - x - x^2}{x^3} = \frac{1}{3}$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

раскр. до 3 степени

$$\sin x = x - \frac{x^3}{6} + o(x^3)$$

$$e^x \sin x = x + x^2 + \frac{x^3}{2} + \frac{x^4}{6} + o(x^4) - \frac{x^3}{6} - \frac{x^4}{6} - \frac{x^5}{12} - \frac{x^6}{24} + o(x^6) + o(x^3) + o(x^4) + o(x^5) + o(x^6) + o(x^6)$$

$$e^x \sin x = x + x^2 + \frac{x^3}{2} - \frac{x^3}{6} + o(x^3) = x + x^2 + \frac{x^3}{3} + o(x^3)$$

примеч:

раскр-го 3 степени

$$e^x = 1 + x + \frac{x^2}{2} + o(x^2)$$

$$\sin x = x - \frac{x^3}{6} + o(x^3)$$

$$e^x \sin x = x + x^2 + \frac{x^3}{2} - \frac{x^3}{6} + o(x^3) = x + x^2 + \frac{x^3}{3} + o(x^3)$$

Пример 6

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} = \lim_{x \rightarrow 0} \frac{\frac{x^3}{3} + o(x^3)}{\frac{x^3}{6} + o(x^3)} = \lim_{x \rightarrow 0} \frac{\frac{1}{3} + o(1)}{\frac{1}{6} + o(1)} = 2$$

$$\sin x = x - \frac{x^3}{6} + o(x^3)$$

$$x - \sin x = \frac{x^3}{6} + o(x^3)$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

$$e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + o(x^3)$$

$$e^x - e^{-x} - 2x = \frac{x^3}{6} - \left(-\frac{x^3}{6}\right) + o(x^3) = \frac{x^3}{3} + o(x^3)$$

Пример 7

$$\lim_{x \rightarrow 0} \frac{\operatorname{tg} x - \sin x}{x^3 + x^4} = \lim_{x \rightarrow 0} \frac{x + \frac{x^3}{3} + o(x^3) - x + \frac{x^3}{6} + o(x^3)}{x^3 + x^4} = \lim_{x \rightarrow 0} \frac{\frac{x^3}{2} + o(x^3)}{x^3 + o(x^4)} \stackrel{|\cdot x^3}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{2} + o(1)}{1 + o(1)} = \left(\frac{1}{2}\right)$$

$$\sin x = x - \frac{x^3}{6} + o(x^3)$$

$$\operatorname{tg} x = \frac{\sin x}{\cos x} = \sin x (\cos x)^{-1} = \left(x - \frac{x^3}{6} + o(x^3)\right) \left(1 - \frac{x^2}{2} + o(x^2)\right)^{-1} = \left(x - \frac{x^3}{6} + o(x^3)\right) \left(1 + \frac{x^2}{2} + o(x^2)\right) = x + \frac{x^3}{2} - \frac{x^3}{6} + o(x^3) = x + \frac{x^3}{3} + o(x^3)$$

т.к.
 $\cos x = 1 - \frac{x^2}{2} + o(x^2) + o(x^2)$

$$(1+x)^q = 1 + qx + \frac{q(q-1)}{2!}x^2 + \frac{q(q-1)(q-2)}{3!}x^3 + o(x^3)$$

$$\left(1 + \left(-\frac{x^2}{2} + o(x^2)\right)\right)^{-1} = 1 + (-1) \left(-\frac{x^2}{2} + o(x^2)\right) + o(x^3) = 1 + \frac{x^2}{2} + o(x^3)$$

Пример 8 $\lim_{x \rightarrow 0} \frac{\ln(1+\operatorname{tg} x) + 1 - e^x + \arcsin x}{x - \sin x}$

Раскладываем до степени, при которой пропадет неопр-ть

$$\sin x = x - \frac{x^3}{6} + o(x^3)$$

$$x - \sin x = \frac{x^3}{6} + o(x^3) \text{ - раскл до 3 степ.inkl.}$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

$$\arcsin x = x + \frac{x^3}{6} + o(x^3)$$

$$(\arcsin x)' = \left(x + \frac{x^3}{6} + o(x^3)\right) \left(x + \frac{x^3}{6} + o(x^3)\right) = x^2 + o(x^3)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3)$$

$$\operatorname{tg} x = x + \frac{x^3}{3} + o(x^3)$$

$$\ln(1+\operatorname{tg} x) = \ln\left(1 + \underbrace{\left(x + \frac{x^3}{3} + o(x^3)\right)}_{\downarrow 0}\right) = \left(x + \frac{x^3}{3} + o(x^3)\right) - \frac{\left(x + \frac{x^3}{3} + o(x^3)\right)^2}{2} + \frac{\left(x + \frac{x^3}{3} + o(x^3)\right)^3}{3} + o(x^3) = x - \frac{x^2}{2} + \frac{2x^3}{3} + o(x^3)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\operatorname{tg} x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

$$\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$$

$$\arctg x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$(1+x)^a = 1 + ax + \frac{a(a-1)}{2!} x^2 + \frac{a(a-1)(a-2)}{3!} x^3 + \dots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+tx) + 1 - e^x + \arcsin^3 x}{x - \sin x} = \lim_{x \rightarrow 0} \frac{x - \frac{x^2}{2} + \frac{2x^3}{3} + o(x^3) + 1 - \left(x + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)\right) + x^3 + o(x^3)}{\frac{x^3}{6} + o(x^3)} =$$

$$= \lim_{x \rightarrow 0} \frac{\frac{x^3}{2} + o(x^3)}{\frac{x^3}{6} + o(x^3)} = 3$$

(N.Y)

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} \operatorname{tg} \frac{\sin \sqrt{2} x}{\sqrt{2}} \right)$$

$$\sqrt{1+x^2} = (1+x^2)^{1/2} = 1 + \frac{1}{2}x^2 + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2}x^4 + o(x^4) = 1 + \frac{x^2}{2} - \frac{x^4}{8} + o(x^4)$$

$$\sqrt{1-x^2} = (1-x^2)^{1/2} = 1 + \frac{1}{2}(-x^2) + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2}(-x^2)^2 + o(x^4) = 1 - \frac{x^2}{2} - \frac{x^4}{8} + o(x^4)$$

$$\sqrt{1+x^2} + \sqrt{1-x^2} - 2 = \left(1 + \frac{x^2}{2} - \frac{x^4}{8} + o(x^4)\right) + \left(1 - \frac{x^2}{2} - \frac{x^4}{8} + o(x^4)\right) - 2 = -\frac{x^4}{4} + o(x^4)$$

$$\sin \sqrt{2} x = \sqrt{2} x - \frac{(\sqrt{2} x)^3}{6} + \frac{(\sqrt{2} x)^5}{120} + o(x^5)$$

$$\frac{\sin \sqrt{2} x}{\sqrt{2}} = x - \frac{x^3}{3} + \frac{x^5}{20} + o(x^5)$$

$$\operatorname{tg} x = x + \frac{x^3}{3} + \frac{2x^5}{15} + o(x^5)$$

$$\operatorname{tg} \frac{\sin \sqrt{2} x}{\sqrt{2}} = \operatorname{tg} \left(x - \frac{x^3}{3} + \frac{x^5}{20} + o(x^5) \right) = \left(x - \frac{x^3}{3} + \frac{x^5}{20} + o(x^5) \right) + \frac{\left(x - \frac{x^3}{3} + \frac{x^5}{20} + o(x^5) \right)^2}{3} + \frac{2 \left(x - \frac{x^3}{3} + \frac{x^5}{20} + o(x^5) \right)^3}{15} + o(x^5)$$

$$\begin{aligned} \left(x - \frac{x^3}{3} + \frac{x^5}{20} + o(x^5) \right)^3 &= x^3 \left(1 - \frac{x^2}{3} + \frac{x^4}{20} + o(x^4) \right) \left(1 - \frac{x^2}{3} + \frac{x^4}{20} + o(x^4) \right) \left(1 - \frac{x^2}{3} + \frac{x^4}{20} + o(x^4) \right) = x^3 \left(1 - \frac{x^2}{3} \right)^3 + o(x^5) \\ &= x^3 \left(1 - 3 \cdot 1 \cdot \frac{x^2}{3} + 3 \cdot 1 \cdot \frac{x^4}{20} - \frac{x^6}{27} \right) = x^3 - x^5 + o(x^5) \end{aligned}$$

$$\operatorname{tg} \frac{\sin \sqrt{2} x}{\sqrt{2}} = \left(x - \frac{x^3}{3} + \frac{x^5}{20} + o(x^5) \right) + \frac{x^3}{3} - \frac{x^5}{3} + o(x^5) + \frac{2x^5}{15} + o(x^5) = x - \frac{x^5}{6} + o(x^5)$$

$$\frac{1}{x} \operatorname{tg} \frac{\sin \sqrt{2} x}{\sqrt{2}} = 1 - \frac{x^4}{6} + o(x^4)$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} \operatorname{tg} \frac{\sin \sqrt{2} x}{\sqrt{2}} \right) = \lim_{x \rightarrow 0} \left(1 - \frac{x^4}{6} + o(x^4) \right) = \left[1 - \frac{0}{6} \right] = 1$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\operatorname{tg} x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

$$\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$$

$$\arctg x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$(1+x)^a = 1 + ax + \frac{a(a-1)}{2!}x^2 + \frac{a(a-1)(a-2)}{3!}x^3 + \dots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} e^{\ln\left(1 - \frac{x^4}{6} + o(x^4)\right)} \\
&= \lim_{x \rightarrow 0} e^{\frac{1}{-\frac{x^4}{6} + o(x^4)}} \\
&= \lim_{x \rightarrow 0} e^{\frac{1}{-\frac{x^4}{6} + o(x^4)} \cdot \boxed{\ln\left(1 - \frac{x^4}{6} + o(x^4)\right)}} \\
&= \lim_{x \rightarrow 0} e^{\frac{-\frac{x^4}{6} + o(x^4)}{-\frac{x^4}{6} + o(x^4)}} = e^{2/3}
\end{aligned}$$

